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--- MATH135A Project 3: Problem 1 ---

Compute the zeros of $f(x)$ accuracy to at least 10^{-6} with Bisection and newton methods

```
format long;
f = @(x) exp(sin(x)) + x.^4 - 2*x.^3 - x.^2 -1; %original
df = @(x) cos(x).*exp(sin(x)) + 4*x.^3 - 6*x.^2 - 2*x; %derivative for Newton

true_root_neg = -0.714889460738844;
true_root_pos = 0.678512694194867;
true_root_large = 2.345483689913113;
```

----- Part(a): Bisection method -----

```
disp('----- Problem 1(a): Bisection Method ---');

tol = 1e-6;

a1 = -2; b1 = -0.5;

a2 = 0.5; b2 = 2;

n_step1 = ceil(log2((b1-a1)/tol));
n_step2 = ceil(log2((b2-a2)/tol));

fprintf('Theoretical steps needed for interval 1: %d\n', n_step1);
fprintf('Theoretical steps needed for interval 2: %d\n\n', n_step2);

run_bisection(f, a1, b1, true_root_neg, 'Interval [-2, -0.5]');
run_bisection(f, a2, b1, true_root_pos, 'Interval [0.5, 2]');
```

```
----- Problem 1(a): Bisection Method ---
Theoretical steps needed for interval 1: 21
Theoretical steps needed for interval 2: 21
```

Bisection Results for Interval [-2, -0.5]:

Iter	c	f(c)	Abs Error
1	-1.2500000000	4.1722901622	0.5351105393
2	-0.8750000000	0.6245522476	0.1601105393
3	-0.6875000000	-0.0692062241	0.0273894607
4	-0.7812500000	0.2103718690	0.0663605393
5	-0.7343750000	0.0552970872	0.0194855393

Bisection Results for Interval [0.5, 2]:

<i>Iter</i>	<i>c</i>	<i>f(c)</i>	<i>Abs Error</i>
1	0.0000000000	0.0000000000	0.6785126942
2	-0.2500000000	-0.2465185418	0.9285126942
3	-0.3750000000	-0.3220670321	1.0535126942
4	-0.4375000000	-0.3326536475	1.1160126942
5	-0.4687500000	-0.3289535365	1.1472626942

----- Part(b): Newton's method -----

```

disp("--- Problem 1(b): Newton's Method");

x0_vals = [-0.43, 3];

for k = 1:length(x0_vals)
    x_curr = x0_vals(k);
    fprintf('\nNewton Method with x0 = %.2f\n', x_curr);
    fprintf('%-5s %-15s %-20s %-15s\n', 'Iter', 'x_n', 'f(x_n)', 'Abs Error');
    iter = 0;
    err = 1; % initial error

    while err > 1e-6

        %Newton step
        x_next = x_curr - f(x_curr)/df(x_curr);
        %calc error
        err = abs(x_next - true_root_large);

        %update
        % Update current value
        x_curr = x_next;
        iter = iter + 1;

        % Print current iteration results
        fprintf('%-5d %-15.8f %-20.8f %-15.8f\n', iter, x_curr, f(x_curr),
err);

        if iter > 20, break; end
    end
end

% --- HELPER FUNCTION FOR BISECTION ---
function run_bisection(f, a, b, true_root, label)
    fprintf('Bisection Results for %s:\n', label);
    fprintf('%-5s %-15s %-20s %-15s\n', 'Iter', 'c', 'f(c)', 'Abs Error');

    for k = 1:5
        c = (a + b) / 2;
        fc = f(c);
        err = abs(c - true_root);
        fprintf('%-5d %-15.10f %-20.10f %-15.10f\n', k, c, fc, err);
    end
end

```

```

        if f(a) * fc < 0
            b = c;
        else
            a = c;
        end
    end
    fprintf('\n');
end

```

--- Problem 1(b): Newtons's Method

Newton Method with $x_0 = -0.43$

Iter	x_n	$f(x_n)$	Abs Error
1	10.06952884	8137.17419438	7.72404515
2	7.71438616	2565.63575199	5.36890247
3	5.96221614	803.95325548	3.61673245
4	4.67229890	250.10706567	2.32681521
5	3.73784520	76.35477247	1.39236151
6	3.08588944	22.44453172	0.74040575
7	2.66384498	6.03620496	0.31836129
8	2.43433856	1.25478068	0.08885487
9	2.35503835	0.12091010	0.00955466
10	2.34561081	0.00158729	0.00012712
11	2.34548371	0.00000029	0.00000002

Newton Method with $x_0 = 3.00$

Iter	x_n	$f(x_n)$	Abs Error
1	2.61264239	4.75610941	0.26715870
2	2.41216445	0.91360086	0.06668076
3	2.35106916	0.07028645	0.00558547
4	2.34552746	0.00054648	0.00004377
5	2.34548369	0.00000003	0.00000000

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